

Practice 4: M-PSK modulation

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https://github.com/santiagoUIS/GNURADIO_COMUIS_2025_2_B1_G7/tree/Practica_4

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Abstract—In modern digital communication systems, phase modulation is a fundamental technique for transmitting information efficiently and robustly over noisy channels. Among the different variants, M-PSK (multiple-phase shift keying) modulation stands out for its ability to represent multiple bits per symbol through variations in the carrier phase. This technique is widely used in wireless, satellite, and data communication standards due to its balance between spectral efficiency and noise resistance. This report presents the implementation and analysis of M-PSK modulation using the GNURadio interface. The practice was carried out using two methods, a VCO (voltage-controlled oscillator)-based approach and a truth table (constellation vector) method. The objective of the experiment was to characterize the modulated signals in the time, frequency, and constellation domains, determine their bandwidth, and evaluate the effect of constellation modifications on performance. The comparison between M-PSK, QPSK, and a custom constellation allowed us to identify how phase configuration influences signal robustness and efficiency.

keywords: *Phase modulation, M-PSK, Domain, Carrier phase, Noise resistance, Bandwidth, Efficiency.*

I INTRODUCTION

Digital communication systems rely heavily on modulation schemes that efficiently encode information onto a carrier signal. *M-PSK* is one of the most widely used phase modulation techniques, where data is represented through discrete phase shifts of a constant-amplitude carrier. The higher the value of M , the more bits are transmitted per symbol, improving spectral efficiency at the cost of reduced noise immunity.

This laboratory exercise focused on the implementation of *M-PSK* in a software-defined radio (SDR) environment using GNURadio. The objective was to understand both the mathematical and practical aspects of phase modulation, including constellation representation, symbol mapping, and power spectral density (PSD).

During the practical session, the relationship between the bit rate and symbol rate, spectral efficiency, and the influence of the number of phases M on the robustness of the system in the face of noise is analyzed. In addition, the impact of the channel on the shape and dispersion of the constellations is studied, observing the progressive degradation of the system with the decrease in SNR[1].

Together, these activities allow us not only to validate the theoretical principles of phase shift keying modulation, but also to develop the ability to implement, analyze, and optimize digital transmission schemes in software-defined radio environments.

II METHODOLOGY

This section presents in an orderly and detailed manner the experiments carried out during the laboratory, including simulations and other considerations addressed within the practice. All experiments were conducted in the GNURadio. The methodology was systematic to ensure reproducibility and integrity within the results obtained.

II-A System overview

The transmitter was modeled in baseband using complex arithmetic blocks[2]. A binary data source was mapped to phase values corresponding to the M-PSK constellation. The output signal was visualized both as a complex envelope and as an RF equivalent through cosine and sine mixing.

The diagram of the system used is illustrated in Fig. 1, which shows the data source, mapping, modulation and analysis sinks. This diagram has been developed based on this reference [video](#), which summarizes the theory necessary to carry out this implementation, including the use of VCO methods and truth tables worked out during the practice.

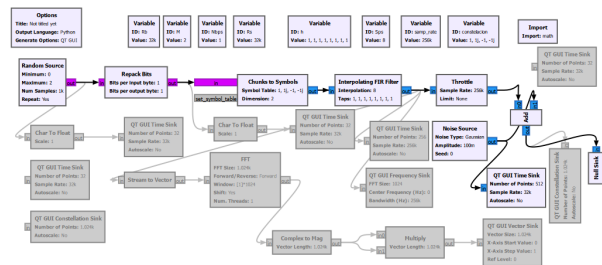


Figure 1. Block diagram for performing M-PSK modulation.

II-B VCO-based implementation

The first method used a Voltage-Controlled Oscillator (VCO) to generate a phase-modulated carrier. A stream of digital bits was converted into voltage levels that adjusted the oscillator's phase. The resulting signal had a constant amplitude but discrete phase changes corresponding to the transmitted symbols.

This approach provided an intuitive understanding of how information can be encoded into phase variations and helped visualize the direct impact of symbol timing and frequency deviation on the waveform.

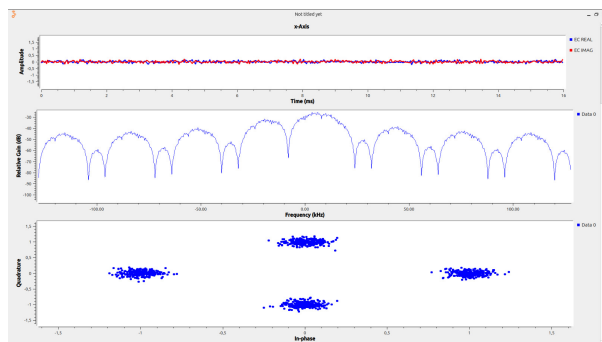


Figure 2. Output of the block diagram for the VCO-based signal modulation.

Figure 2 shows the time-domain waveform, power spectral density, and constellation diagram of the VCO-based Q-PSK transmitter, where the constant amplitude and four discrete phase states confirm the correct generation of a quadrature phase-shift keyed signal.

II-C Truth-table method (constellation vector)

The second method directly defined the M-PSK constellation through a complex vector where each element represented a point on the unit circle. The block "Chunks to Symbols" mapped groups of bits to these points, effectively producing the baseband complex envelope. Fig. 3 shows the vector configuration.

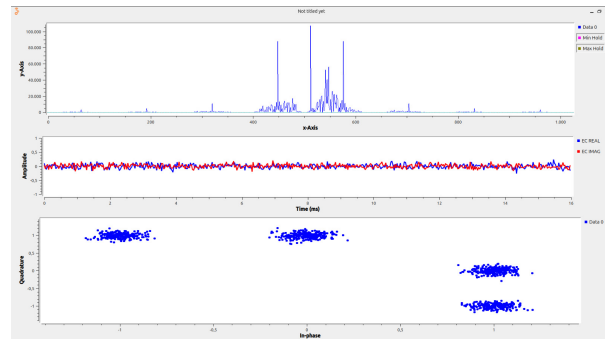


Figure 3. Output of the block diagram for the M-PSK modulation using the bit-to-symbol mapping defined by the truth table.

This method provided a clearer and more flexible approach to visualize and modify the constellation geometry compared to the VCO-based implementation.

II-D Measurement procedure

- The time-domain waveform was analyzed to verify the correct phase transitions for each symbol.
- The PSD of the complex envelope was obtained to determine bandwidth and identify null frequencies associated with the symbol rate.
- Constellation diagrams were observed under noise-free and noisy conditions to evaluate symbol clustering.
- Custom constellations were implemented to observe how phase rotation and spacing impact performance.

III EXPERIMENTS AND RESULTS

This section presents the experimental evidence, figures, and analyses obtained from the GNURadio simulations.

III-A Modified constellations

Some modifications were proposed to the original constellation (see Fig. 2) to change its nature. First, the spacing between the points was altered, losing the orthogonality between the points of the constellation, and then a 45° rotation was performed on that reference, as shown in Fig. 4. Both modifications demonstrated how inadequate phase spacing can cause group overlap and increase the probability of symbol error. This is precisely because orthogonality is lost and the noise present can generate interference between linearly dependent coordinates. However, in the case of reference phase change or rotation, the average symbol energy does not change, nor does the signal-to-noise ratio. This is because the magnitude of the symbols is preserved.

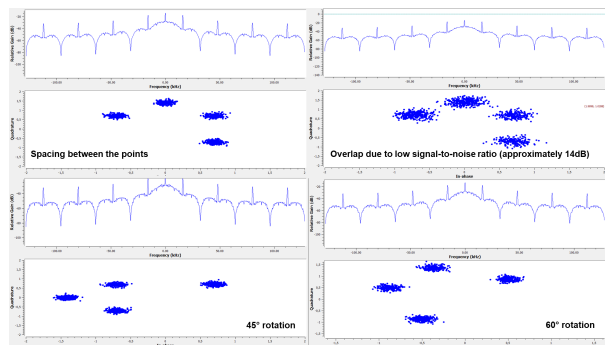


Figure 4. Experiments varying phase and position of points on the constellation diagram.

Subsequently, the noise amplitude was increased to verify whether there was interference between symbols. This amplitude was increased from 0.1[v] to 0.2[v]. In the same way, a 60° phase shift was tested to identify changes in the diagram, which revealed changes in the positions of the constellation points, but always complying with the standard that is equal to the amplitude of the modulated signal.

III-B Complex envelope bandwidth

The bandwidth of the complex envelope was determined from the PSD analysis shown in Fig. 2. The main lobe width defines the effective transmission bandwidth, which depends on the symbol rate and pulse shaping filter. The measured bandwidth was approximately 18kHz, consistent with theoretical expectations for a 10ksp/s symbol rate and roll-off factor close to 0.35.

III-C Power Spectral Density and zeros

The PSD shown in Fig. 5 exhibits nulls at integer multiples of the symbol rate, confirming the theoretical relationship between time-domain symbol transitions and frequency-domain periodicity. These nulls indicate that energy is concentrated within a well-defined bandwidth, making M-PSK spectrally efficient.

The PSD follows a sinc^2 pattern, which arises when a signal is formed by symbols transmitted at a finite symbol rate (R_s).

The spectral nulls occur theoretically at:

$$f = n \cdot R_s, \quad n \in \mathbb{Z} \quad (1)$$

IV CONCLUSIONS

The implementation and analysis of M-PSK modulation allowed for a deeper understanding of phase modulation principles and their impact on system performance. Through the experiments, it was verified that the

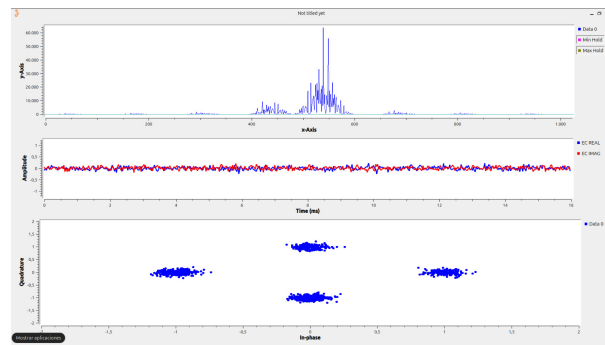


Figure 5. Power Spectral Density (PSD) showing spectral zeros related to symbol rate.

constellation diagram directly reflects the phase states defined by the modulation order M , and that any modification to these phases alters symbol mapping and error probability.

The results demonstrated that the complex envelope representation provides an efficient way to study modulated signals at baseband, removing the carrier while preserving all essential information regarding amplitude and phase. Observations from the Power Spectral Density confirmed that the energy distribution and spectral zeros correspond closely to the theoretical symbol rate, validating the relationship between time-domain symbol transitions and frequency-domain characteristics.

The bandwidth measurements obtained experimentally were consistent with theoretical predictions, confirming that M-PSK schemes maintain spectral efficiency while transmitting multiple bits per symbol. However, as M increases, the phase separation between adjacent constellation points decreases, making the system more susceptible to noise and distortion. This trade-off between spectral efficiency and robustness was clearly visible in the constellation diagrams and PSD plots.

References

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